HARDWARE-BASED FLIGHT SAFETY SYSTEM

DroneGuard

Intelligent Drone Health Monitoring & Mission Optimization System

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01

Problem Statement

Autonomous drones fail silently — until it's too late.

No real-time health visibility

Motor, ESC, battery and propeller wear go undetected mid-flight, with no continuous telemetry-based diagnosis.

Unpredictable in-flight failures

Overheating motors, vibration anomalies or GPS drift are discovered only after a crash or mission abort.

Static, one-size-fits-all missions

Flight speed, altitude and routes are pre-set without accounting for current drone condition or payload.

Battery mismanagement

Missions are attempted without verifying if remaining charge and health can safely complete the round trip.

02

Proposed Solution

Two integrated engines working from the same live telemetry stream.



Health Monitoring Module

- Tracks motor temperature, vibration & current draw
- Monitors battery voltage, health & discharge trend
- Checks propeller balance, IMU drift and GPS quality
- Flags anomalies before, during and after each flight



Mission Optimization Engine

- Combines drone health with payload & route data
- Recommends safe speed, altitude & waypoint order
- Sets a dynamic return-to-home battery threshold
- Estimates flight time before takeoff is authorized

03

Methodology

A continuous sense → assess → decide → act loop for every mission.



1

Sensor Data Acquisition

Onboard sensors stream motor, ESC, battery, IMU, GPS & comm telemetry



2

Parameter Analysis

Temperature, vibration, current, voltage & signal quality are processed



3

Health Assessment

Component-wise and overall drone health score is computed



4

Mission Evaluation

Health score is matched against payload, distance & conditions



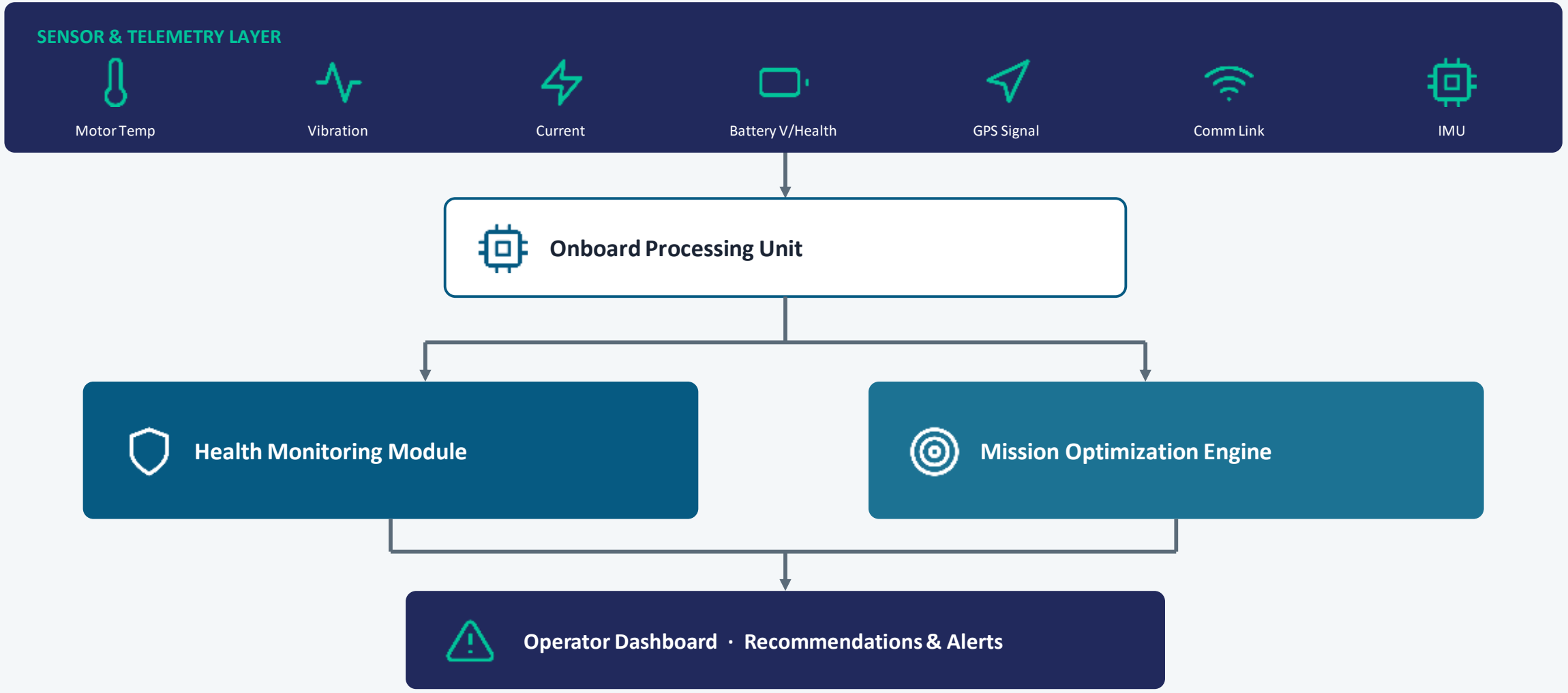
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Recommendation / Alert

Optimized parameters are issued, or the operator is alerted

04

System Architecture



05

Expected Outcomes

Safer autonomous flight through continuous awareness and adaptive planning.

Fewer Failures

Early detection of motor, ESC & battery anomalies prevents in-flight incidents

Efficient Flights

Speed, altitude & routes tuned to real drone condition and payload

Reliable RTH

Dynamic return-to-home threshold avoids battery-depletion crashes

Timely Alerts

Operators are warned instantly when a mission isn't safe to fly

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